

Problem Statement

Inconsistent treatment and **inefficient use of hospital resources** arise from the current skeletal traction device:

1. The **traction frame is fixed** to a hospital bed, which impacts the ability to perform necessary scans to develop accurate treatment plans. It also requires a special hospital bed, which is expensive and can take time to bring to the patient.
2. **Hanging weights** (around 20 pounds) do not apply consistent traction, introduce risk of injury to surrounding people, and requires a staff member to hold the weight during patient transport.
3. The geometry of the solution causes the **patient to slide down the bed**, which also leads to inadequate traction and need for staff to frequently readjust the patient.

Needs Statement

Orthopedic surgeons need a way to apply skeletal traction to **adult patients with femoral shaft fractures** that **maintains correct femur length** throughout treatment despite changes in patient position to **prevent femur displacement** that can cause further damage.

User Needs

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| 1. Device shall apply traction to femur fracture per current treatment requirements | 7. Device shall provide line of sight and access to the pin site |
| 2. The device shall maintain the set length and position of femur throughout treatment | 8. Device shall be biocompatible |
| 3. User must be able to know the force applied | 9. Device shall be sterilizable |
| 4. Device shall be radiolucent for x-rays | 10. Device packaging, and labeling shall be consistent with FDA standards |
| 5. The device must be compatible with use of most hospital bed, including X-ray beds | 11. Device shall be composed of reusable components |
| 6. The pin must align with current traction pin use and safety standards for distal femoral pins or proximal tibial pins | |

